Chathuranga Weeraddana

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1. Objectives

My work spans **distributed optimization, distributed machine learning, signal processing, and wireless communications**, with a focus on the **theoretical and algorithmic foundations of learning and optimization for large-scale networked systems**. I aim to advance **rigorous methodological frameworks** while applying these approaches to **data-driven and interdisciplinary challenges**. Alongside my research, I am committed to **high-quality teaching, curriculum development, and student supervision**, integrating research-led insights into teaching to support student learning and engagement in AI and data-driven systems.

2. Research Interests

Distributed optimization, distributed machine learning, multi-stage decision problems, surrogate modeling in optimization, signal processing, wireless communication.

3. Teaching Interests

Including, but not limited to: Mathematical optimization, machine learning, statistical signal processing.

4. Degrees

University of Oulu **Finland**
Doctor of Science, Telecommunication Engineering *2011*
Thesis: Optimization Techniques for Radio Resource Management in Wireless Communication Networks
Reviewers: Prof. Leandros Tassiulas (Yale University, USA), Prof. Lars K. Rasmussen (KTH, Sweden)
Opponents: Prof. Lars K. Rasmussen (KTH, Sweden), Prof. Olav Tirkkonen (Aalto University, Finland)

Asian Institute of Technology **Thailand**
Master of Engineering, Telecommunication Engineering *2007*
Thesis: Performance Analysis of OFDM Systems with Random Residual Frequency Offset

University of Moratuwa **Sri Lanka**
Bachelor of Science, Electronic and Telecommunication Engineering *2004*

5. Leadership Experiences

- **DESIRE6G** (*Deep Programmability and Secure Distributed Intelligence for Real-Time End-to-End 6G Networks*), 2023-2025, funded under Horizon Europe / SNS JU
  - Managed the University of Oulu's contributions to the Horizon Europe DESIRE6G, coordinating technical activities, deliverables, and collaboration with consortium partners.
- **IEEE MERCon**, the flagship international conference of the University of Moratuwa, Sri Lanka
  - International Relations Chair, 2022
  - Publication Chair, 2020
- **NCTM, National Conference on Technology and Management**, organized by the Research Center, SLIIT, Sri Lanka
  - Co-Chair, 2016

6. Research Funding Applications

Research Council of Finland **2025**
Status *Under Review*
Title: Scalable Perception and Adaptive Reinforcement for Multi-Agent Decisions over Communication-Limited Networks (SPARNet)
Role: Co-PI
Amount: 0.35 M€ of 0.9 M€

NSF Program: Verticals-enabling Intelligent Network Systems (VINES) **2025**
Status *Under Review*
Title: Semantic-Aware Human–Robot Collaboration for Personalized Content Delivery in Scalable, Re- silient VR/AR Systems (SPARCS)
Role: Co-PI
Amount: 0.35 M€ of 0.75 M€ from Research Council of Finland

Research Council of Finland **2025**
Status *Not Funded*
Title: Engineering MARL Solutions as an Enabler for AI Native Communication Networks (EMENAITIVE ComNet)
Role: PI
Amount: 0.85 M€

7. Research and Professional Experience

CWC-RT, University of Oulu **Finland**
Docent *2024/09 - To Date*
Working in the subgroup Intelligent Connectivity and Networks/Systems Group (ICON)

CWC-RT, University of Oulu **Finland**
Senior Research Fellow *2022/08 - 2024/09*

Dept. of Electronic & Telecommun. Engineering, University of Moratuwa **Sri Lanka**
Senior Lecturer (equivalent to Associate Prof.) *2018 - 2022/07*

Dept. of Electrical & Computer Engineering, SLIIT **Sri Lanka**
Senior Lecturer (equivalent to Associate Prof.) *2015 - 2018*

Dept. of Automatic Control, KTH Royal Institute of Technology **Sweden**
ACCESS Postdoc Researcher *2012-2014*

8. Selected Publications

1. H. Abeynanda, **P. C. Weeraddana**, and C. Fischione, "On the Characteristics of the Conjugate Function Enabling Effective Dual Decomposition Methods," *IEEE Transactions on Signal Processing*, vol. 74, pp. 572-588, Jan. 2026.

2. T. AlShammari, **P. C. Weeraddana**, and M. Bennis, "BayGO: Decentralized Bayesian Learning and Information-Aware Graph Optimization Framework," *IEEE Transactions on Signal Processing*, vol. 72, pp. 2101-2116, Apr. 2024.

3. S. Ranaweera, **P. C. Weeraddana**, P. Dharmawansa, and C. Fischione, "On the Convergence of Inexact Gradient Descent With Controlled Synchronization Steps," *IEEE Signal Processing Letters*, vol. 30, pp. 703-707, May. 2023.

4. H. Abeynanda, **P. C. Weeraddana**, G. H. J. Lanel, and C. Fischione, "On the Primal Feasibility in Dual Decomposition Methods Under Additive and Bounded Errors," *IEEE Transactions on Signal Processing*, vol. 71, pp. 655-669, Feb. 2023.

5. M. Ding, S. Atapattu, **P. C. Weeraddana**, and J. Evans, "Energy-Efficiency Fairness of Interference Multi-Relay Networks for Multi-User Communication," *IEEE Communication Letters*, vol. 25, no. 3, Mar. 2021.

6. S. Magnússon, **P. C. Weeraddana**, M. Rabbat, and C. Fischione, "On the convergence of alternating direction lagrangian methods for nonconvex structured optimization problems," *IEEE Transactions on Control of Network Systems*, vol. 3, no. 3, pp. 296–309, Oct. 2016.

7. S. Magnússon, **P. C. Weeraddana**, and C. Fischione, "A distributed approach for the optimal power flow problem based on admm and sequential convex approximations," *IEEE Transactions on Control of Network Systems*, vol. 2, no. 3, pp. 238–253, Oct. 2015.

8. G. Athanasiou, **P. C. Weeraddana**, C. Fischione, and L. Tassiulas, "Optimizing client association for load balancing and fairness in millimeter-wave wireless networks," *IEEE/ACM Transactions on Networking*, vol. 23, no. 3, pp. 836-850, Mar. 2014.

9. **P. C. Weeraddana**, M. Codreanu, M. Latva-aho, A. Ephremides, and Carlo Fischione, "Weighted sum-rate maximization in wireless networks: A review," *Foundations and Trends in Networking, Now Publishers Inc.*, vol. 6, no. 1-2, pp. 1–163, 2012.

10. **P. C. Weeraddana**, M. Codreanu, M. Latva-aho, and A. Ephremides, "Weighted sum-rate maximization for a set of interfering links via branch and bound," *IEEE Transactions on Signal Processing*, vol. 59, no. 8, pp. 3977–3996, Aug. 2011.

11. **P. C. Weeraddana**, M. Codreanu, M. Latva-aho, and A. Ephremides, "Multicell downlink weighted sum-rate maximization: A distributed approach," *IEEE Transactions Signal Processing*, vol. 61, no. 3, pp. 556–570, February 2013.

9. Research Supervision (Selected)

- **Hansi Abeynanda**; Doctoral student; School of Electrical Engineering and Computer Science, KTH, Sweden
  - Thesis: Distributed Machine Learning Over Networks
  - My supervision period: 2023-To Date
  - Outputs: **IEEE SP (2026, see Sec. 8 item 1); IEEE TCNS (under preparation 2026)**
- **Tamara AlShammari**; Doctoral student; Faculty of Information Technology and Electrical Engineering, University of Oulu, Finland
  - Thesis: TBD
  - My supervision period: 2022-To Date
  - Outputs: **IEEE SP (2024, see Sec. 8 item 2); JMLR (under preparation 2026)**
- **Hebatalla Issa**; Doctoral student; Faculty of Information Technology and Electrical Engineering, University of Oulu, Finland
  - Thesis: TBD
  - My supervision period: 2024-To Date
  - Outputs: **ICML (submitted 2026); IEEE TCCN (under preparation 2026)**
- **Sandushan Ranaweera**; Bachelor's student; University of Moratuwa, Sri Lanka
  - My supervision period: 2021-2022
  - Outputs: **IEEE SP Letter (2023, see Sec. 8 item 3)**
- **Hansi Abeynanda**; Master's student; University of Sri Jayewardenepura, Sri Lanka
  - Thesis: Distributed Consensus with Dual Decomposition Methods Under Additive and Bounded Errors
  - My supervision period: 2021-2022
  - Outputs: **IEEE SP (2023, see Sec. 8 item 4)**

- **Sindri Magnússon**; Doctoral student; School of Electrical Engineering and Computer Science, KTH, Sweden
  - My supervision period: 2014
  - Outputs: **IEEE TCNS (2015, see Sec 8 item 6)**
- **Sindri Magnússon**; Master's student; School of Electrical Engineering and Computer Science, KTH, Sweden
  - My supervision period: 2013
  - Outputs: **IEEE TCNS (2015, see Sec. 8 item 7)**

10. Teaching and Curriculum Development

Courses Taught

Div. of Network and Systems Engineering, KTH Royal Institute of Technology **Sweden**
Lecturer, Graduate Course *2022/03-2022/06*
MM Optimization Algorithms with Machine Learning Applications
`https://chathugithub1.github.io/Home/courses/courseinfo.html`

Faculty of Engineering, University of Moratuwa **Sri Lanka**
Lecturer, Graduate Cousrses *2018-2022*
Engineering Decision Theory, Broadband Communications, Digital Communications, Advanced Instrumentation

Faculty of Engineering, University of Moratuwa **Sri Lanka**
Lecturer, Undergraduate Courses *2018-2022*
Introduction to Telecommunication, Teletraffic Engineering

Dept. of Electrical & Electronics Engineering, University of Peradeniya **Sri Lanka**
Lecturer, Graduate Course *2018- 2022*
Optimization in Communication Systems

Dept. of Electronic Engineering, Yangon Technical University **Myanmar**
Lecturer, Graduate Course *2018 - 2019*
Introduction to Mathematical Optimization

Dept. of Electrical & Computer Engineering, SLIIT **Sri Lanka**
Lecturer, Graduate Courses *2015 - 2018*
Several Communication and Signal Processing Courses

Dept. of Electrical & Computer Engineering, Curtin University **Australia**
Lecturer, Undergraduate Courses through SLIIT *2015 - 2018*
Several Communication and Signal Processing Courses

Dept. of Automatic Control, KTH Royal Institute of Technology **Sweden**
Lecturer, Graduate Courses *2013-2014*
Network Optimization, Principles of Wireless Sensor Networks

Centre for Wireless Communications, University of Oulu **Finland**
Lecturer, Graduate Courses *2010*
Convex Optimization with Applications in Wireless Communications and Networks

Developing New Teaching Material (Graduate Level)

Did bear the responsibilities of developing courses

- MM Optimization Algorithms with Machine Learning Applications (sole responsibility, Digital Futures 'Scholar-in-Residence' program, KTH Royal Institute of Technology, Sweden, 2022)
- Optimization in Communication Systems (sole responsibility, Dept. of Electrical & Electronics Engineering, University of Peradeniya, Sri Lanka, 2018)

- Introduction to Mathematical Optimization (sole responsibility, Dept. of Electronic Engineering, Yangon Technical University, Myanmar, 2018)
- Network Optimization (partial responsibility, Dept. of Automatic Control, KTH Royal Institute of Technology, Sweden, 2013)

Developing New Teaching Material (Undergraduate Level)

Did bear the responsibilities of developing courses

- Introduction to Engineering Optimization (sole responsibility, Faculty of Eng., University of Moratuwa, Sri Lanka, 2020)
- Convex Engineering Design (sole responsibility, Faculty of Eng., University of Moratuwa, Sri Lanka, 2020)
- Network Traffic Engineering (partial responsibility, Faculty of Eng., University of Moratuwa, Sri Lanka, 2020)

11. Invited Talks and Visits for Research/Teaching

KTH Royal Institute of Technology — **Sweden**
Guest Researchers — *2022/03/01 - 2022/06/01*

Dept. of Electrical & Electronics Engineering, University of Peradeniya — **Sri Lanka**
Visiting Lecturer — *2018-2022*

Dept. of Electronic Engineering, Yangon Technical University — **Myanmar**
Visiting Lecturer — *2019/08/19 - 2019/09/01*

Dept. of Electronic Engineering, Yangon Technical University — **Myanmar**
Visiting Lecturer — *2018/09/10 - 2018/09/20*

16th International Conference on Industrial and Information Systems (IEEE ICIIS) — **Sri Lanka**
Guest Speaker — *2021*
Session: 5G Evolution and 6G

Faculty of Graduate Studies, University of Moratuwa — **Sri Lanka**
Resource Person — *2022/08/23*
Event: Research Lounge

12. Awards and Honours

- **ACCESS Post-Doctoral Fellowship:** Among 1 of the 3 post-doctoral fellows selected from ACCESS Linnaeus Centre, KTH Royal Institute of Technology, Sweden, 2012
- **Best doctoral thesis award:** Dept. of Comm. Eng., University of Oulu, Finland, 2011
- **Second best paper award:** Asilomar Conference on Signals, Systems and Circuits 2010
- **GETA grant:** Graduate School in Electronics, Telecommunications and Automation grant for a 4 years of graduate school position in University of Oulu, Finland, 2011
- **Students' research grants:** Tauno Tönning Säätiö, Finland, 2011, 2009
- **Students' research grants:** Oulun Yliopiston Tuksäätiö, Finland, 2010
- **Students' research grants:** Nokia Foundation, Finland, 2010, 2009
- **Students' research grants:** Tekniikan Edistämissäätiö Stiftelsen, Finland, 2008
- **The AIT alumni association prize:** in recognition of the most outstanding academic performance among students in the School of Engineering and Technology, AIT, Thailand, 2007
- **The Sonera prize:** in recognition of the most outstanding academic performance among students in Telecommunication, AIT, Thailand, 2007

- **King's Scholarship:** from His Majesty King Bhumibol Adulyadej of the kingdom of Thailand, 2005

13. University Administrative Experience

CWC-RT, University of Oulu — **Finland**
Member, M.Sc. Admission Evaluation Committee — *2025 – To Date*
Member, Follow-up group for doctoral students — *2023 – To Date*

University of Moratuwa — **Sri Lanka**
Member, Quality Assurance Committee — *2019 – 2022*
Member, Engineering Research Unit — *2018 – 2021*

Dept. of Electronic and Telecommunication Engineering, University of Moratuwa — **Sri Lanka**
The Academic Coordinator — *2021, 2022*

Faculty of Engineering Technology, The Open University of Sri Lanka — **Sri Lanka**
Senior Consultant — *2021, 2022*

Faculty of Engineering, SLIIT — **Sri Lanka**
Member, Curriculum Development Committee — *2016*
Member, Accreditation Committee — *2016*

14. Other Key Academic Merits

- **Exemplary Reviewer:** Certificate of appreciation from IEEE Communication Society, IEEE Wireless Communications Letters, 2020,2021
- **Reviewer:** IEEE Transactions of Communications, IEEE Wireless Communications Letter, IEEE Transactions of Signal Processing, and IEEE Transactions of Wireless Communications, among others
- Completed an Induction Training course at the Staff Development Centre, Open University of Sri Lanka (2019), covering key pedagogical frameworks including outcomes-based education (OBE), constructive alignment, scaffolding, constructivism, curriculum design and mapping, program design, and assessment methods.

Referees

Professor Carlo Fischione
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